Machine Learning for Bioengineering

Prof. Stefano Diciotti's Module – Practical part

Second Cycle Degree/Two-Year Master's in Biomedical Engineering

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Setting up the development platform (Google Colab)

1. Go to eol.unibo.it, log in to today's session, and download the file **'Template_ML4Bioengineering_20260622.ipynb'.**
2. Go to https://colab.research.google.com/ , and log in at the top right with your Google account.
3. Upload the template to Google Colab, for example, by selecting **'File -> Upload Notebook -> Upload'**

Documentation

Directly use the following links for the Python documentation (DO NOT use Google or other search engines because of firewall settings):

- https://pandas.pydata.org/docs/
- https://scikit-learn.org/stable/
- https://numpy.org/doc/
- https://matplotlib.org/stable/index.html

Exercise: Logistic regression and hyperparameter optimization

A synthetic biomedical dataset has already been generated and is available in the variables X and y.

Split the dataset into training (60%), validation (20%), and test (20%) sets.

Train a Logistic regression model using the following values of C:

C_values = [0.001, 0.01, 0.1, 1, 10, 100]

For each value of C, train the model on the training set and evaluate its accuracy on the validation set. Select the value of C that achieves the highest validation accuracy.

Once the best value of C has been identified, retrain the model on the union of the training and validation sets and evaluate its accuracy on the test set.

At the end of the notebook, report:

- the validation accuracy obtained for each value of C;
- the selected value of C;
- the final test accuracy.

In a separate text cell, answer the following questions:

1. As C increases, the complexity of the logistic regression model increases. What can you say about the model capacity? How is this related to underfitting and overfitting?

2. In this exercise we used a nested hold-out procedure (training, validation, and test sets) rather than a nested cross-validation. Why is the nested hold-out approach appropriate in this case?

Submitting the solution on eol.unibo.it

Save your Python code on your PC, rename it, and upload it to eol.unibo.it